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SUPERSEDING
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MILITARY SPECIFICATION

COMMUNICATION SYSTEMS FOR TRAINING DEVICES GENERAL SPECIFICATION FOR

This specification has been approved by the Naval Training Equipment Center, Department of the Navy

1. SCOPE

1.1 Scope. This general specification establishes the requirements for standardized four wire voice communication systems required in training devices (see 6.2.1)

1.2 Classification. The communication systems used in training devices are classified as follows:

Type I - Instructor/trainee communications including device maintenance communication provisions.

Type II - Communication systems which simulate tactical or non-tactical communication systems or subsystems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-23991	Training Devices, Military, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to. Naval Training Equipment Center, Orlando, Florida 32813, Attn: Code N-212, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 69GP

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STANDARDS

Military

MIL-STD-130	Identification Markings of U. S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials, parts and processes. Unless otherwise specified in the detail specification (see 6.2.2), materials, parts and processes shall be in accordance with MIL-T-23991.

3.2 Design. The trainer design shall be in accordance with MIL-T-23991, the detail specification, and as specified herein.

3.2.1 Mechanical.

3.2.1.1 Construction. The communications stations and consoles shall be of modular construction such that installation assembly and disassembly can be accomplished without special tools. Chassis and subassemblies shall have provisions for securing and locking in place for shipment.

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3.2.1.2 Strength. The communications equipment shall be capable of withstanding, without damage, stresses incident to movement, handling intransit, hoisting, tie down aboard transporting vehicles and final installation and operation.

3.2.1.3 Size. The size of all units of the communications equipment, including maintenance parts and auxiliary equipment shall be designed to meet the requirements of the detail specification.

3.2.2 Electrical and electronic.

3.2.2.1 Design features. The design of all types of communications systems shall incorporate the following design features:

(a) Systems shall be designed for independent transmit and receive (4-wire) operations.

(b) Input and output circuits to and from communication stations shall be balanced.

(c) Input and output impedances of communications station audio circuits shall be 600 ohms ± 10 percent, unless otherwise specified by the detail specification.

(d) Automatic Gain Controlled (AGC)(see 6.2.3) amplifiers shall be used to amplify and distribute the voice circuits to the circuit subscribers.

(e) Solid-state circuitry shall be used in all preamplifiers, speaker amplifiers, AGC amplifiers, and switching circuits.

(f) All electronic subassemblies, such as, preamplifiers, speaker amplifiers, and AGC amplifiers, shall be designed as plug-in modules.

(g) Subassembly items, such as, preamplifiers and speaker amplifiers, which may be used in different types of stations shall be identical in design.

(h) Audio cables shall be used to interconnect communication stations to central distribution equipment. All conductors in the cable shall be individually shielded twisted pairs interwoven to minimize crosstalk coupling. The cables shall contain a ground shield and overall polyvinyl jacket. Cable conductors shall be made from stranded wire; cable conductor size shall be a minimum of 22 gauge (AWG).

(i) Headset and microphone cables shall be of sufficient length to allow operators and observers to stand.

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(j) All electronic modules, such as, preamplifiers, speaker amplifiers, and AGC amplifiers, shall be designed to operate on the same voltage which shall be regulated ±10 percent.

(k) When required by the detail specification, emergency power shall provide continuous operation of the communication system for a period as specified following the initial failure of primary power. A relay system shall be utilized to provide automatic switching from primary power to emergency power with automatic reset to primary power.

3.2.2.2 Communication capabilities. The design of four wire voice communication systems shall provide the following communication capabilities:

(a) Switching facilities to select any assigned channel one at a time and to communicate on this channel utilizing an operator's headset/handset instrument or an integral combination microphone/speaker.

(b) Facilities to indicate when channels are busy.

(c) The capability to monitor selectively any combination of assigned channels on an integral speaker amplifier at stations, or to monitor only the assigned circuit(s) at the stations specified in the detail specification.

(d) Panel-mounted headset/handset receptacles.

(e) Facilities to permit using both carbon and dynamic microphones. (An optional plug-in microphone preamplifier shall be used with headsets which contain dynamic microphones.)

(f) Separate volume controls for headset/handset listening and speaker monitoring of circuits.

(g) Facilities to mute the speaker when transmitting on the same channel to which the speaker is assigned.

(h) Front-panel access to plug-in subassemblies, such as, preamplifiers, speaker amplifiers and other plug-in subassemblies.

(i) Cable connector receptacles and mating connector plug for cables which enter and leave the station.

(j) Panel-mounted power ON-OFF switch and indicator lamp.

(k) Maintenance and support communications networks by allocating

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a separate conference circuit of the intercommunications system. Each maintenance and support station shall provide a switch to select operation in either a two-way speaker/microphone or a headset (with extention cord) mode. Selection of one mode shall disable the other.

3.2.2.3 Central distribution cabinet(s). Central distribution cabinet(s) shall be provided for the communications subsystems to terminate and to distribute the communications circuits of these subsystems. The central distribution cabinets shall contain the following facilities:

(a) Plug-in AGC amplifiers.

(b) Termination for the cables from each communications station in the system.

(c) Test panel for patching, testing, and monitoring each individual audio circuit to and from assigned communications stations.

(d) Transmission test set, including AC voltmeter and audio signal generator, for test and maintenance purposes.

(e) Fuse and alarm panel containing circuit protection (fuses) for all stations and common equipment and shall be equipped with an audible alarm circuit with cutoff switch and, a visual alarm lamp which remains illuminated until the blown fuse is replaced.

(f) Regulated power supply which supplies the operating voltages to all electronic modules in the communication system. The AC input to the DC power supply shall be 115 VAC, ± 10 percent, 60 hertz, single phase. For installations, such as shipboard equipment, the voltage input to the DC supply shall be as specified in the detail specification.

(g) AC control panel with circuit breaker and POWER ON indicator lamp.

(h) Sliding shelf to facilitate maintenance and test functions.

(i) Storage drawer for patchcords, test leads, spares, etc.

3.2.2.4 Instructor and special console communications stations. The electrical design of instructor and other special console communications stations shall provide the following communications capabilities in addition to those specified in 3.2.2.2

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(a) Facilities for selectively accessing and communicating on any circuit in the overall communications system, such as, intercom, radiophone, sound powered phone, and support communications circuits, using a headset or handset instrument.

(b) Facilities for connecting any audio circuit to the input of an external audio recorder.

(c) Capability for instructor to override all other transmissions.

(d) Headset/handset connector receptacle(s) and extension jacks for mounting under the shelf of the console.

3.2.2.5 Operator communications subsystems. The electrical design of the operator four wire communications stations shall be compatible with the functional operation requirements of the detail specifications for the subject training device.

3.2.2.6 Sound-powered telephone subsystem. Training devices which simulate sound-powered telephone systems shall utilize operational type equipment facilities and, except where nonstandard switching or distribution facilities are required, no special designs are necessary for implementing sound-powered telephone communications subsystems.

3.3 Reliability. Reliability shall be in accordance with the detail specification.

3.4 Maintainability. Maintainability shall be in accordance with the detail specification.

3.5 Transportability. Transportability shall be in accordance with the detail specification.

3.6 Performance. Electrical performance of the communication system shall be as follows.

(a) System frequency response: Flat ± 2dB (decibel) from 300 to 3300 Hz referred to a nominal system level of 0 dBm (dB above or below one milliwatt across 600 ohms) at 1 kHz (excluding headset microphone transmitter and receiver elements).

(b) Noise level: At least 50 dB below nominal signal level of 0 dBm at 1 kHz.

(c) Crosstalk level: At least 50 dB below nominal signal level from 300 to 3300 Hz.

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(d) Total harmonic distortion: Less than 5 percent at nominal system level of 0dBm

(e) System level: 3 dBm (1 milliwatt across 600 ohms at 1000 Hz), nominal (receive level at stations)

(f) System level stability: Constant within 3 dB from 1 to 100 station loads

(g) Speaker output power (at stations): Four watts maximum, at 1000 Hz, adjustable

(h) Speaker output power (Maintenance and support communications network). Four watts at 1000 Hz, adjustable

(i) Headset levels (at stations): 0 dBm maximum, -20 dBm minimum, continuously adjustable

(j) Transmit levels (from stations): -10 dBm nominal.

3.7 Details of components. The communications systems shall consist of the equipment and facilities necessary to implement the voice communications networks used in training devices. Unless otherwise specified in the detail specification, the systems shall include the following major equipment components:

(a) Operator's (see 6.2.4) communications stations for use on intracommunications, intercommunications, sound-powered phone, and device maintenance and support voice communications network.

(b) Instructor's and other special console communications stations for use on intracommunications, intercommunications, sound-powered phone, and device maintenance and support voice communications networks

(c) Central equipment cabinet(s) to house distribution amplifiers, DC power supplies, patching, and termination facilities, test facilities, level indicators, and other common equipment as required for each of the associated individual types of voice communications networks

(d) Headsets, handsets, and sound-powered telephone instruments for use with communications stations assigned to the voice communications networks

(e) Extension headset jacks for mounting under the shelf of consoles, equipped with foot-operated, push-to-talk switches

(f) Interconnecting cable and connector assemblies for all communications stations and associated equipment items

(g) Emergency power provision.

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3.8 Electromagnetic interference suppression. Electromagnetic interference (EMI) suppression of the subsystem shall be in accordance with the detail specifications. In the event the detail specification has no EMI suppression requirements, the subsystem shall be designed to meet the conducted emission (CE) and conducted susceptibility (CS) requirements of MIL-STD-461.

3.9 Dimensions. Dimensions and tolerances shall be in accordance with the detail specification.

3.10 Weight. Weight shall be in accordance with the detail specification.

3.11 Color. The selection of color shall be in accordance with the detail specification and MIL-T-23991.

3.12 Finish. Finish shall be in accordance with MIL-T-23991, and the detail specification.

3.13 Training device nameplates. Unless otherwise specified in the detail specification, nameplates shall be in accordance with MIL-T-23991.

3.14 Government-furnished property (GFP). Government-furnished property, shall be handled as specified in the detail specification.

3.15 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. Quality assurnace provisions shall be in accordance with MIL-T-23991 and the detail specification.

4.2 Reliability demonstration. A reliability demonstration shall be conducted in accordance with the detail specification.

4.3 Maintainability demonstration. A maintainability demonstration shall be conducted in accordance with the detail specification.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the detail specification.

6. NOTES

6.1 Intended use. The intended use of the communications system defined in this specification is for communications systems which are a part of various training devices. The ultimate objective of this specification is to define and specify standard system design techniques, electrical performance requirements, and subassembly items where applicable

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for use in implementing the communications facilities of different types of training devices. The quantities and types of stations, their detailed layouts and operational requirements, are specified in the detail specifications of the training device.

6.2 Definitions. Definitions are provided in MIL-T-23991 and as follows herein.

6.2.1 General specification. A general specification is a specification for a type or class of training device and covers general requirements for such training devices beyond those cited in MIL-T-23991. In addition, the general specification covers areas of design, performance, and tests that are peculiar to a given class of training devices.

6.2.2 Detail specification. A detail specification is a specification which defines the details of design, performance, and tests that are not specified in general specification. As used in this specification, it refers to the detail specification of the parent training device of which the communications equipment is considered a subassembly

6.2.3 AGC amplifier. Automatic Gain Controlled (AGC) or compression-type amplifiers are specified to minimize differences in circuit levels due to variances in speaking levels of operators.

6.2.4 Operator. As used herein, "Operator" refers to the student undergoing training in a training device satellite position. However, the term "Operator" is used for the broader application where the satellite station is not necessarily manned by a student.

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